College of Engineering Trivandrum

Department of Computer Applications

20MCA203 . Design and Analysis of Algorithms (Semester 3)

Assignment II

1.Differentiate Deterministic algorithms and Randomised algorithms with an example each.

2. Write the Randomized Quick Sort algorithm. Explain how randomized pivot selection differs from the conventional Quick Sort algorithm.

3. For the array $[24, 8, 42, 75, 29, 17, 3]$, illustrate one possible execution of Randomized Quick Sort by selecting a random pivot and showing the first partitioning step.

4. Explain the Schwartz-Zippel Lemma. How this Lemma can be used to test the identity of two polynomials